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ISO standards for terminology resources management

Are they FAIR enough?

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The ‘FAIR terminology’ paradigm is the first initiative that aims to establish a strong interrelation between the FAIR (Findability, Accessibility, Interoperability, and Reusability) principles, promoted at the European level to foster the Open Science movement, and the specifications advocated in the framework of ISO TC 37/SC 3 about the management of terminology resources. Through the complementary adoption of the SC 3 standards, the paradigm, therefore, proposes guidelines paving the way for the FAIRification of terminological data, metadata, and infrastructure. In this context, this study has a threefold objective. First, we will show the impact of three ISO TC 37/SC 3 standards (mainly ISO 16642: 2017; ISO 12620: 2019 and ISO 30042: 2019) for the foundation of the FAIR terminology paradigm. Secondly, we will present the integration of this data representation model in the framework of the European project ‘Terminology Without Borders’ (TWB) launched in 2019 by the Terminology Coordination Unit (TermCoord) of the European Parliament. Finally, we intend to reflect in a critical perspective to highlight some possible limitations of the previously mentioned SC 3 standards in terms of FAIRification of terminological data.

Keywords: FAIR terminology, FAIR principles, terminology management, ISO standards, TermBase eXchange, open science

1. Introduction

Language data is nowadays pivotal in developing modern language technologies – for example, language models like ChatGPT¹ that require vast amounts of data to

1. <https://github.com/NL2G/ChatGPTReview>

be trained. The organisation of such an amount of data in a structured way is fundamental to bringing these technologies into effective action. In this sense, some researchers have discussed the possibility of formally considering structured language data as a high-value data category at the international level, like the European Union Open Data Directive² which will then be transposed into national legislation.

Yet, a lot of language data is not reused; hence, its potential remains unexplored. This is why organisations and institutions have made large investments to support and foster data modeling standardisation and reuse of multilingual data (Smal et al. 2020). An important initiative in this direction is provided by the European Language Resource Coordination (ELRC),³ launched in April 2015 to collect relevant language data for public services. The ELRC-SHARE repository⁴ is used for documenting, storing, browsing, and accessing Language Resources that are collected through the ELRC and considered useful for the community. ELRC-SHARE has language data provided by governmental agencies, administrations and other (public) institutions that possess a collection of mono- or multilingual structured data that are relevant for overcoming the language barriers among European institutions and between European institutions and citizens.

The interinstitutional terminology database InterActive Terminology for Europe (IATE)⁵ of the European Union is another important example of the effort that the EU has put into the preparation of a platform that can face future challenges, including interoperability, modularity, scalability, and data exchange, among other things (Zorrilla-Agut, and Fontenelle 2019).

In this context, standardisation and harmonisation of work in terminology are extremely important to ensure the interoperability and reusability of language resources (Fišer, and Witt 2022). The ISO standards developed within TC 37/SC 3⁶ are intended to provide guidelines for the optimal and consistent management of terminology resources. In particular, the three ISO standards we will mainly refer to in this paper are:

1. ISO 16642: 2017 *Computer applications in terminology – Terminological markup framework* which defines the *Terminological Markup Framework* (TMF) metamodel for the representation of terminological data collections.

2. <https://data.gov.ie/pages/open-data-directive>

3. <https://www.lr-coordination.eu/>

4. <https://elrc-share.eu/>

5. <https://iate.europa.eu/>

6. <https://www.iso.org/committee/48136.html>

2. ISO 12620: 2019 *Computer applications in terminology – Data categories* (now superseded by ISO 12620-1: 2022 *Management of terminology resources – Data categories – Part 1: Specifications* and ISO 12620-2: 2022 *Management of terminology resources – Data categories – Part 2: Repositories*)⁷ which consistently defines the properties of data categories (such as their names and definitions) and their documentation in an open repository.
3. ISO 30042: 2019, *Management of terminology resources – TermBase eXchange (TBX)*, which defines the *TermBase eXchange (TBX)* representation format specifically designed for the exchange of multilingual terminological data.

The TMF marks a crucial point in the succession of terminology standards with a twofold aim (Romary 2001; Romary et al. 2006): It is a meta-model for terminological data representation aiming at facilitating the description and comparison of existing interchange formats, and it provides a mechanism for the flexible definition of interchange formats while safeguarding interoperability between them. TMF is important because it constitutes a meta-structure for the encoding of terminology, it also provides the foundation for the more recent TermBase eXchange (TBX) format (Melby 2015; Pernes et al. 2017). In this respect, TBX defines a family of formats that share a common structure and a limited range of information types, the terminological data categories and their respective specifications that are given in the ISO 12620 Data Category standard. The main purpose of the TBX exchange format is to ensure interoperability and reusability across different software applications (Fišer, and Witt 2022).

These specifications have had an impact in terms of terminological data, metadata and infrastructure curation which is reflected both on the corporate side and on the academic one. Focusing on the academic perspective, we describe in this paper the first initiative, named ‘FAIR terminology’, that fosters the adoption of the three aforementioned TC 37/SC 3 standards to meet the needs of ‘FAIRness’ of research data promoted at the European level within the Open Science movement.⁸

In this context, this study has a threefold objective. First, we show the impact of the three ISO TC 37/SC 3 standards for the foundation of the FAIR terminology

7. It is important to mention that in this paper we refer to the ISO 12620 version of 2019 although currently withdrawn and replaced by the two 2022 versions of the standard above mentioned. This is because the FAIR terminology paradigm that is the object of this study was developed in the 2017–2020 time frame taking into consideration the ISO standards in force at the time. An adaptation to the specifications of the last two standards is currently being implemented.

8. https://research-and-innovation.ec.europa.eu/strategy/strategy-2020-2024/our-digital-future/open-science_en

paradigm. Secondly, we present the adoption of the FAIR terminology paradigm in the framework of TWB promoted by the Terminology Coordination Unit (TermCoord)⁹ of the European Parliament. The TWB project aims to provide multilingual terminological data collections in different specialised domains for open re-use by terminologists, specialised translators, and interpreters. The FAIR terminology paradigm was integrated as the reference data modeling for the project, and this also led to the development of a new terminology management system, called 'FAIRterm' (Vezzani 2021), as a web application for supporting the TWB terminology workflow and the export of data in TBX format for their reuse in different CAT tools. Thirdly, we intend to reflect in a critical perspective to highlight some possible limitations of the previously mentioned ISO standards in terms of FAIRification of terminological data. Our analysis focuses on two aspects: (1) the type of identifiers used within any TBX concept entries (ISO 30042: 2019); and (2) the need to guarantee both the syntactic and semantic interoperability (Decker et al. 2000) of this kind of resource.

The paper is organised as follows: In Section 2, we analyse a review of the impact of the TC/37 ISO standards on academic research and projects that use these standards, as well as the need for the opening to the Linguistic Linked Open Data (LLOD) world. In Section 3, we present the core aspects of our proposal related to FAIR terminology: the impact on the TWB project, and the open issues of the main ISO standards for terminology. Section 4 concludes our proposal with some final remarks and future works.

2. Impact of terminological standards on research

In this section, we analyse the impact of the terminology standards on academic research concerning terminology and, more in general, tools for the management of terminology and semantic interoperability. We present a review of the most recent research papers about the design and implementation of terminological tools that comply with the above-mentioned ISO TC/37 standards or works that discuss the advantages and drawbacks of the use of such standards in the development of this kind of software.

We performed a keyword search and a manual screening of the results on Google Scholar using the keywords 'terminology' together with one of the three main ISO standards for the management of terminology resources: ISO 30042, ISO 16642, and ISO 12620. We filtered the results by the publication date, and we

9. <https://termcoord.eu/>

included papers from 2017 (the year of the ‘oldest’ ISO update among the three, the ISO 16642:2017) until 2023.

We divide the presentation of these works into two parts: (1) the papers that use or discuss these standards (Section 2.1), and (2) the papers that present the approaches and the issues related to the transformation of the XML-based terminological databases into RDF-based LLOD (Section 2.2).

2.1 Use of standards

The standards are very important, as they allow resources or tools to be used interchangeably. The writing of such documents is a complex task, as the resulting standard should be able to allow for “repurposability” (Warburton 2018, 119). Indeed, a terminology resource that can support a multitude of users and usages is deemed to be multipurpose and repurposable. Moreover, it must be able to seamlessly integrate into various computing environments and software applications. In this respect, Salgado et al. (2019) present a study of an interoperable lexical database that aims at dictionary content reusability. In particular, the main task is updating the first complete dictionary of the Academia das Ciências de Lisboa (DACL), and the entries of this dictionary use the attributes `xml:id` and `xml:lang` in compliance with ISO 16642 to encode the basic elements of the dictionary.

The questions related to interoperability are also the focus of Vacalopoulou et al. (2019). In this work, the authors discuss the use of standards for a controlled terminology of Greek museums. In particular, they claim that an adequate choice of a standardised set of data categories for the classification of cultural heritage would result in better interoperability between different systems used in different museums. At the same time, different domains require different data categories: A government-sponsored terminological database documenting a nation’s official languages will require a different description compared to a corporate termbase designed to support global marketing. The use of the correct data categories in a termbase is the focus of the review on the development of a Data Category Repository for language resources and the related ISO 12620 presented by Warburton and Wright (2020). In particular, the authors discuss the introduction in this standard of the possibility to select only the data categories needed for a specific termbase.

The use of standards to organise terminological data can also be the source of new approaches and solutions for other research areas, such as information retrieval. In this context, Benoît (2020) explores whether the linguistic registers in ISO 12620 can improve the findability of medical resources. Cultural heritage and health, two of the domains previously mentioned, are two examples of situations where the creation of a terminological resource for a specific domain can be

an added value for the development of information service platforms. Arndt and Runnwerth (2021) address how terminological resources should be brought under a common modeling paradigm with the use of standards for the design of knowledge organisation systems. In particular, the authors present a set of best practices that use a concept-oriented approach with or without the support of TBX to establish a controlled vocabulary, in the sense of *thesaurus* (Arndt and Runnwerth 2021, 169). In the same year, Mihăescu (2021) explored the possibility of envisaging a conceptual hierarchy for the organisation of a termbase by introducing information concerning the relations of a concept with other concepts. This work aims to combine and constitute terminological knowledge-based banks following the principles of the TMF standard.

To ensure interoperability between different termbases in the structure and exchange with external terminology producers and consumers, Fišer and Witt (2022) present the Norwegian terminology portal, Termportalen,¹⁰ for a wide range of subject areas. This resource is an example of the use of TBX as the primary exchange format in the Common Language Resources and Technology Infrastructure (CLARIN)¹¹ framework. In addition, Meisinger et al. (2022) describe the CLARIN Concept Registry¹² and the refinement of the related ISO 12620:2019 to target only terminological databases rather than providing data categories in a more general sense.

Finally, if we look at the corporate side, Warburton (2021) illustrates how the adoption of the three above-mentioned TC 37/SC 3 standards (and more generally ISO TC 37/SC 1 standards) is an integral part of the daily work of a ‘corporate terminologist’. Furthermore, these standards not only concern the professional figures of the terminologist or specialised translator (Galinski, and Weissinger 2010), but they are also relevant for acquiring the skills for more recent (or rather recently renewed) professional figures such as those of the ‘technical communicator’. While at the ISO level, the international standard ISO/DIS 24183 *Technical Communication – Vocabulary* is being defined, at the Italian national level, for instance, the UNI¹³ 11483: 2021 standard *Attività professionali non regolamentate – Comunicatore professionale – Requisiti di conoscenza, abilità e autonomia e responsabilità* [Non-regulated professions – Communication Manager – Knowledge, skill, autonomy and responsibility requirements] was recently published to define, among others, the skills and knowledge expected for the role of the tech-

10. <https://www.termportalen.no/>

11. <https://www.clarin.eu/>

12. The CLARIN Concept Registry is the successor of ISOcat (the ISO TC37 Data Category Registry (DCR) created in 2009).

13. Ente Nazionale Italiano di Unificazione (Italian National Unification).

nical communicator. This professional figure has the task of defining, creating, and distributing information for the safe, effective, and efficient use of products in the broadest sense (technical systems, software, services, etc.). Within UNI 11483: 2021– and to corroborate the impact that the above-mentioned TC 37/SC 3 standards have in the corporate context – explicit reference is made to ISO 16642: 2017; ISO 12620: 2019 and ISO 30042: 2019 which are listed among the knowledge and know-how requirements of a technical communicator.

2.2 From TBX to RDF

The use of standards is a fundamental step for the reusability of digital resources. Linking different terminological datasets to each other would contribute to the creation of a seamless ecosystem of terminologies that can be easily accessed and navigated and creates added value by allowing applications to access and exploit a network of linked terminologies.

Nevertheless, the TBX standard is not immediately usable when a researcher wants to expose the terminological data in a different format such as Linked Data, particularly in the world of LLOD (Di Nunzio, and Vezzani 2021). TBX was not designed to link data on the web but rather as a means of ensuring the exchange, and consequently, the reuse of data.

In this regard, numerous studies have been conducted for combining TBX-native models and the Semantic Web by means of the Resource Description Framework (RDF) W3C standard model for data interchange on the Web. Cimiano et al. (2015) described the seminal work, named TBX2RDF,¹⁴ to apply linked data principles to terminological datasets and present an approach to transform termbases in TBX format to RDF. Their approach was based on the *lemon* model,¹⁵ an RDF model developed to support publishing lexical resources as linked data, and a series of best practices to transform terminologies from TBX into the linked data format.

In Speranza et al. (2020) and Di Buono et al. (2020), the authors describe the process and the methodologies adopted in the conversion of linguistic resources encoded in TBX (the previous version of the standard dated 2008) together with their semantic enrichment based on the formalisation of ontological information into terminologies. Piccini et al. (2023); Bellandi (2023), and Bellandi et al. (2023) present from different points of view (diachronic study of medical language, digital humanities, FAIR terminology) a tool to favor a deep synergy between the resources in TBX and the resources in RDF. The authors propose a tool to auto-

14. <http://tbx2rdf.lider-project.eu/converter/tbx2rdf.html>

15. <https://www.w3.org/2016/05/ontolex/>

matically convert the resources created in the most recent version of TBX in RDF (Ontolex-Lemon). Chozas and Declerck (2022) take a different approach to the problem of porting the multilingual terminology resources onto a Linked Data compliant representation language. In particular, the authors propose an extension module for OntoLex-Lemon that deals with the representation of terminological data that is not covered in the core module.

An alternative point of view that focuses more on the interrelationships between conceptual levels is presented by Reineke and Romary (2019). The authors focus on the interrelation between TBX and the Simple Knowledge Organization System (SKOS)¹⁶ which are both concept-oriented applications that should interoperate more closely. They propose a mapping of a part of the vocabulary listed in the SKOS reference and the terminological data categories, and their respective specifications, from the data category repository DatCatInfo.¹⁷ A last example that draws attention to the problem of the conversion between TBX and RDF from a different perspective is given by Di Nunzio and Vezzani (2021). In this work, the authors take a step back and look at these issues as a data modeling problems. The hypothesis is that if the terminological data are modeled correctly at an abstract level, then the data themselves can be exported or exposed in any format and maintain both the linguistic dimension and its relationship to the conceptual dimension.

3. A FAIR terminology: Uses and issues

As outlined by Wilkinson et al. (2016), the optimal management of research data is not a goal in itself, but rather it is the key that leads to the enhancement of knowledge and scientific research. This is the theoretical premise underlying the development of the FAIR principles, a set of guidelines for producing and disseminating data promoted at the European level to foster the Open Science movement.¹⁸ In particular:

1. *Findability* implies that data and metadata are uniquely and persistently identified by assigning PIDs (Persistent Identifiers). In addition, the data must be fully described using metadata which must, in turn, explicitly indicate what data they refer to. Finally, both elements must be indexed on findable resources.

16. <https://www.w3.org/2004/02/skos/>

17. <https://datcatinfo.net/>

18. <https://www.go-fair.org>

2. *Accessibility* implies that data and metadata can be retrieved using their identifier using standard, open, free, universally applicable communication protocols and, if necessary, providing authentication/authorisation procedures. Finally, metadata must be accessible even when the data is no longer available.
3. *Interoperability* implies that data and metadata use a formal, accessible, and shared knowledge representation language. In addition, these two elements must use vocabularies respecting the FAIR principles and include, where appropriate, references to other data and metadata.
4. *Reusability* implies that the data and metadata are exhaustively described with relevant and precise attributes, that they are issued with licenses for use and that they have clearly indicated the details concerning their origin. Finally, both elements should follow the benchmark standards for the area of interest.

In this panorama, Vezzani (2022) proposes an adaptation of the FAIR principles to the management of data and metadata produced in the research area of terminology. The FAIR terminology¹⁹ paradigm was developed to provide a digital terminology workflow and a data structure that respects the needs for findability, accessibility, interoperability, and reusability. This paradigm relies upon the complementary application of the three above-mentioned ISO TC/37 SC3 standards: ISO 16642: 2017; ISO 12620: 2019, and ISO 30042: 2019 represent the first step to be taken for a multilingual terminology resource to be FAIR.

Given their intrinsic objectives, these three standards have partial overlap with all four FAIR principles, especially for the interoperability and reusability of terminological resources. However, for each standard, we identified a prominent characteristic to guide the design of a FAIR termbase. In particular, ISO 16642: 2017 – which defines the Terminological Markup Framework (TMF) – presents a modular approach for the design of interoperable terminological data collections, and it has one clear objective that is to “[...] facilitate cooperation and to prevent duplicate work, [...] as well as for sharing and exchanging data” (ISO 16642: 2017, vi). This standard is therefore more oriented towards ‘interoperability’. The terminological data collections as defined in ISO 16642: 2017 can collect various types of information, called data categories, that need to be identified and stored in different environments. ISO 12620: 2019 (now superseded by ISO 12620-1: 2022 and ISO 12620-2: 2022)²⁰ is dedicated to the specification of a “mechanism for creating, documenting, harmonising and maintaining data category specifications in a data category repository” (ISO 12620: 2019, 1). In particular, this standard describes how a data category should be identified for effective use and how it

19. <https://purl.org/fairterm>

20. Please see footnote 7.

should be accessed. In this sense, ISO 12620: 2019 is more oriented towards the ‘findability’ and ‘accessibility’ of a resource. Lastly, ISO 30042: 2019 describes the TermBase eXchange (TBX) format, as a framework for representing structured terminological data in XML. This standard is “[...] designed to support various types of processes involving terminological data, including analysis, descriptive representation, dissemination, and exchange in various computer environments” (ISO 30042: 2019, vi). In this sense, the main objective of this standard is to facilitate the exchange of terminological data, even for different purposes, making this standard more oriented towards the ‘reusability’ aspects of a FAIR resource.

The combination of these standards constitutes the realisation of the FAIR terminology paradigm. This paradigm was first conceived during the design and implementation of the multilingual and multipurpose medical terminology resource TriMED²¹ (Vezzani et al. 2018; Vezzani and Di Nunzio 2020a–b). It was later fully adopted in the development of two other multilingual terminological resources: CAMEO – *CommerciAl terMinology rEsOurce* – for the international trade sector (Vezzani and Di Nunzio 2022) and DITTO – *Disarmament International Treaty TerminOlogy* – for the domain of international disarmament (Vezzani et al. 2022).

The FAIR terminology paradigm – and consequently the adoption of the three standards – has had a significant impact on the framework of the European project TWB promoted by the Terminology Coordination Unit of the European Parliament.

3.1 Terminology without borders

In 2019, the Terminology Coordination Unit (TemCoord) of the European Parliament launched the TWB²² initiative which reflects and supports the objective of the Directorate-General for Translation (DG TRAD)²³ of the European Parliament to communicate with citizens using clear and unambiguous language. As mentioned on the website of the initiative: “[t]he main aim of the project is to enhance communication in several domains by tailoring terminology to citizens’ needs. This aim is achieved by using several multilingual projects.”²⁴

The objective of promoting clear and comprehensible terminology even for citizens who are not experts in a specialised domain is pursued in several het-

21. <http://purl.org/trimed>

22. <https://yourterm.eu>

23. <https://www.europarl.europa.eu/translation/en/introduction>

24. https://ec.europa.eu/education/knowledge-centre-interpretation/news/terminology-without-borders-o_en

erogeneous fields. In particular, the domains of specialisation refer to the field of (1) medicine (YourTerm MED),²⁵ (2) environment (YourTerm ENVI),²⁶ (3) culture (YourTerm CULT),²⁷ (4) protection of women (YourTerm FEM),²⁸ (5) justice (YourTerm JURI),²⁹ (6) maritime (YourTerm MARE),³⁰ (7) education (YourTerm EDU),³¹ (8) agri-food (YourTerm FOOD),³² (9) technology (YourTerm TECH),³³ and finally, (10) economy and finance (YourTerm FIN).³⁴

Within each of these domains, more specific sub-projects are proposed that address different subdomains. For example, within the framework of the YourTerm MED project, seven sub-projects are currently available and related to, among others, bipolar disorders, alternative and complementary medicine, and vaccines.

Each of these projects contains collections of multilingual terminological data which are developed by different partners who actively collaborate with the Terminology Coordination Unit to make these terminological products freely accessible to the community. The partners can be both expert-terminologists working individually, and groups of students coordinated by professors at the universities affiliated with this initiative. In general, the workflow includes the following steps:

1. Compilation of specialised corpora;
2. Terminology extraction and identification of relevant terms;
3. Compilation of multilingual terminology records;
4. Revision and validation of the terminological data;
5. Publication of the terminology collection on the project website;
6. Integration on the IATE database.

The main guiding principle of the Terminology Coordination Unit initiative is based on the criterion of flexibility in the management of these terminology projects. Indeed, the various partners can decide independently to work on their specialised corpus – subject to approval by the Unit – or to join existing projects and enrich the terminology of a given collection from a multilingual perspective. In

25. <https://yourterm.eu/yourterm-med/>

26. <https://yourterm.eu/yourterm-envi/>

27. <https://yourterm.eu/yourterm-cult/>

28. <https://yourterm.eu/yourterm-fem/>

29. <https://yourterm.eu/yourterm-juri/>

30. <https://yourterm.eu/yourterm-mare/>

31. <https://yourterm.eu/yourterm-edu/>

32. <https://yourterm.eu/yourterm-food/>

33. <https://yourterm.eu/yourterm-tech/>

34. <https://yourterm.eu/yourterm-fin/>

this regard, the ToolBox³⁵ section of the website offers a list of suggested tools to complete the workflow, such as Sketch Engine³⁶ (Kilgarrieff et al. 2014) for compiling the corpus, and TermoStat³⁷ (Drouin 2003) for terminology extraction. Regarding the compilation of multilingual terminological records step, the tool currently used is the FAIRterm³⁸ web application developed in 2021 (Vezzani 2021) in collaboration with the Department of Linguistics and Literary Studies, and the Department of Information Engineering of the University of Padua. The idea of developing this tool stems from a series of reflections that emerged during the meetings of the steering committee of the TWB initiative. In particular, the needs expressed concerned:

1. The definition of a set of shared data categories to be included in the terminological record offered to partners;
2. The adoption of a format for the terminological record so that the terminological data can be exported and reused, for example, in computer-assisted translation tools.

Concerning this second point, one of the main difficulties at the beginning was to find a solution for the use of heterogeneous, and sometimes non-structured, format for the representation of terminological records (such as records stored in Excel or Word). In fact, despite being very useful tools for the rapid collection of data and notes, these formats require time-consuming manual work to guarantee high data quality and homogeneity across different files. Starting from these requirements, we have combined the initiative promoted by the research group of the University of Padua aimed at the optimal and uniform structuring of terminological research data into the FAIRterm web application. This application intends to meet the FAIR terminology paradigm by implementing the three ISO TC 37 SC 3 standards for a better harmonisation of the terminology workflow within the TWB projects.

We currently have a total of 586 users from 27 different countries in the world.³⁹ Around 38% of the users use FAIRterm in the framework of the TWB project and these are mainly master students, PhD students and professors at the

35. <https://yourterm.eu/toolbox/>

36. <https://www.sketchengine.eu/>

37. <http://termostat.ling.umontreal.ca/>

38. <https://shiny.dei.unipd.it/fairterm/>

39. The complete list of countries is: Argentina, Austria, Belgium, Canada, China, Czech Republic, France, Georgia, Greece, Ireland, Italy, Latvija, Lietuva, Lithuania, Luxembourg, Nederland, Perú, Poland, Portugal, Romania, Russia, Spain, Sweden, Switzerland, Ukraine, United Kingdom, United States.

partner universities affiliated with the project. The remaining 62% are users who are not directly involved in the TWB project and who are using the application for professional purposes such as corporate terminologists, technical communicators, technical writers, interpreters, and specialised translators.

3.2 Is FAIR terminology FAIR enough?

In the previous sections, we have described the FAIR principles and their applications to the design and implementation of terminological databases. In this section, we highlight our perspective on the open issues of this paradigm and the possible future directions in terms of improving the ISO standards from a FAIR principles point of view.

3.2.1 *How difficult is it to implement FAIR terminology?*

The implementation of a FAIR terminological database means, in simple terms, complying with the three ISO standards described in the previous sections. But how difficult is it to implement a database of such type? Our experience in designing and implementing a termbase fully compliant with these standards tells us that, besides the complexity of the terminological record structure, one of the main challenges resides in the choice of data categories. If a required data category is not listed in the DatCatInfo Repository,⁴⁰ ISO 12620: 2019 gives all the details about how to implement the necessary repository for these new data categories. In practice, this means preparing a Web server with the compulsory requirement. In many cases, this part would be a major overkill for a terminological project. This is partially confirmed by the list of tools that claim to support TBX⁴¹ (very few) provided by the official TBX website.

Indeed, we have searched⁴² for terminological databases that are currently using TBX (at least to export the data) to see how many of these databases correctly comply with the standard. The current situation is quite disappointing from a FAIR point of view: most of the termbases are either not compliant with the latest version of ISO 30042: 2019 (some of them still refer to ISO 30042: 2008) or do not even consider the TBX format. The first notable example is the EU IATE

40. DatCatInfo (<https://datcatinfo.net/>) is the Data Category Repository that contains data category specifications that are used in standards, such as ISO 30042 (TBX).

41. <https://www.tbxinfo.net/tbx-support/>

42. Google Search performed on 2023/03/17, the query is “terminology database” without quotes.

termbase that declares a kind of extension of the TBX Core module⁴³ without describing the data category ‘reliabilityCode’ that is used to indicate the reliability of the term in the entry. The UNTERM⁴⁴ database, the official terminology portal of the United Nations, allows the user to search for terms but not to export data. The only available resource is the official list of countries that can be saved as an Excel file. The AGROVOC,⁴⁵ a multilingual controlled vocabulary covering all areas of interest of the Food and Agriculture Organization of the United Nations, provides a LLOD set about agriculture available for public use. It only allows browsing and downloading data in formats different from TBX. TERMDAT,⁴⁶ the terminology database of the Federal Administration of Switzerland, allows browsing but not downloading, while JURIVOC,⁴⁷ the trilingual thesaurus of the Swiss Federal Supreme Court, allows downloading Word files with the glossaries. The WIPO⁴⁸ Portal, the official platform of the World Intellectual Property Organization, allows the browsing of the records and the conceptual maps only. The Microsoft language⁴⁹ portal allows the download of the TBX multilingual termbase, but the encoding refers to the previous ISO 30042: 2008. The same happens with the NATO terminology database⁵⁰ that is encoded in the old TBX standard.

3.2.2 *Is anything FAIR missing in the ISO standards?*

Being compliant with the ISO standards is an important step for paving the way for a FAIR resource. However, strictly following all the guidelines in the ISO documents may not always result in a completely FAIR approach. We refer in particular to the ISO 30042: 2019 and the ‘findability’ property that, from a FAIR perspective, means that identifiers must meet the following two conditions: be globally unique and be persistent. At present, however, the identifiers of the concept entries in TBX are rather unique only at the local level (of the same TBX instance, see Section 8.5.3 of ISO 30042: 2019) since “[t]he id and target attributes work together to point unambiguously between elements in the same TBX document instance” (ISO 30042: 2019, 18). This means that the standard requires

43. In the XML header: `<?xml-model href="./TBXcoreStructVo3_TBX-IATE_integrated.rng" type="application/xml" schematypens="http://relaxng.org/ns/structure/1.0"?>`

44. <https://unterm.un.org/>

45. <https://agrovoc.fao.org/>

46. <https://www.termdat.ch/search>

47. <https://www.bger.ch/ext/jurivoc/live/en/jurivoc/Jurivoc.jsp>

48. <https://patentscope.wipo.int/>

49. <https://www.microsoft.com/en-us/language>

50. <https://nso.nato.int/natoterm/>

that concept entries are identifiable within the database and not on a global level. This mismatch in the type of identification – local for concept entries vs. global for FAIR principles – results in two problems: the findability (in FAIR terms) of a concept entry is not guaranteed, and, consequently, the interoperability between different terminological resources is less feasible.

This is an important point because, not only it give us the possibility to suggest an improvement to the ISO standard, but it also leads to a reflection on the obstacles in the transformation strategies of the TBX data model –an XML-based format – into a data model compliant with the needs of representation of LLOD in the Semantic Web digital ecosystem.

3.2.3 *How about semantic interoperability?*

The representation of knowledge for the correct data processing, sharing, and reuse between applications is mainly based on two requirements: syntactic interoperability⁵¹ and semantic interoperability⁵² (Decker et al. 2000). The former refers to describing the exact format of the information to be exchanged such that applications can read the data and get a representation that can be exploited. Syntactic interoperability is high when software needed to manipulate data is readily available (such as XML parsers). The latter is about the ability to transmit data with unambiguous, shared meaning. This is accomplished by adding data about the data (metadata) and linking each data element to a controlled, shared vocabulary. For example, agreements on reference data, in the form of taxonomies, controlled vocabularies, or ontologies are key prerequisites for achieving semantic interoperability.

On the one hand, the TBX standard, being an XML-based terminology exchange format, fulfils the syntactic interoperability requirement and is designed to make terminology databases easier to maintain, distribute, and use. On the other hand, the same TBX standard is not immediately useful when a researcher wants to expose the data structured as Linked Data (Piccini et al. 2023). Attempts to fill this semantic gap have been proposed in the literature (Bellandi et al. 2023; Di Nunzio and Vezzani 2021). The most important initiative is the *Guidelines for Linguistic Linked Data Generation: Multilingual Terminologies (TBX)*⁵³ which describes best practices for transforming multilingual terminologies, particularly those available in TBX format, into a Linked Data version by means of a set of recommended vocabularies to be used in doing this conversion and describes the structure of the resulting RDF. However, these guidelines lack some fundamental

51. <https://codata.org/rdm-terminology/syntactic-interoperability/>

52. <https://codata.org/rdm-terminology/semantic-interoperability/>

53. <https://www.w3.org/2015/09/bpmlod-reports/multilingual-terminologies/>

points in the interpretation of the data categories and the theoretical aspects that lie behind the terminological database and that cannot be immediately mapped into a lexical-oriented resource (Bellandi et al. 2023). For this reason, we propose a more careful approach that requires an explicit declaration, by the user who wants to transform the data, about the mapping between data categories and vocabularies.

4. Conclusions and future work

In this paper, we have discussed the impact of the three ISO TC 37/SC 3 standards on current research on terminology and, in particular, as the foundation of the FAIR terminology paradigm. We have presented the adoption of this paradigm in the framework of TWB by means of the FAIRterm web application. This application supports an efficient terminology workflow and the export of data in TBX format for their reuse in different CAT tools. Finally, we have presented a critical analysis of some limitations of the ISO standards in terms of the FAIRification of terminological data. As future work, we propose an interactive mapping from TBX to the LLOD world where the user must explicitly declare the mapping between data categories and LLOD vocabularies to better align these data models.

Funding

This work has been carried out within the framework of the Centro Studi in Terminologia Computazionale (CENTRICO) of the University of Padova (Italy) and supported by the Portuguese national funding through the FCT – Portuguese Foundation for Science and Technology, I.P. as part of the project UIDB/LIN/03213/2020 and UIDP/LIN/03213/2020 – Linguistics Research Centre of NOVA University Lisbon (CLUNL).

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Publication history

Date received: 30 March 2023

Date accepted: 31 August 2023